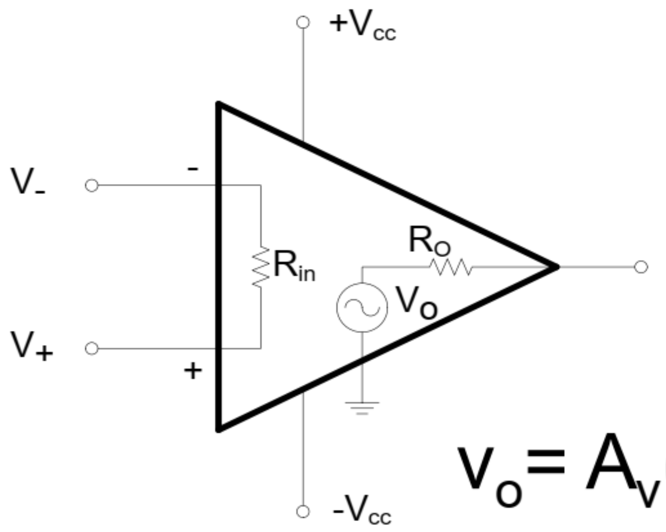
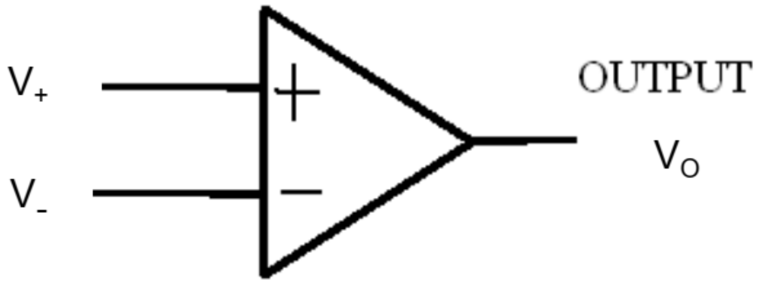
Operational Amplifier
(Op- Amp)

Op-Amp Model



$$v_o = A_V(v_+ - v_-)$$



Symbol

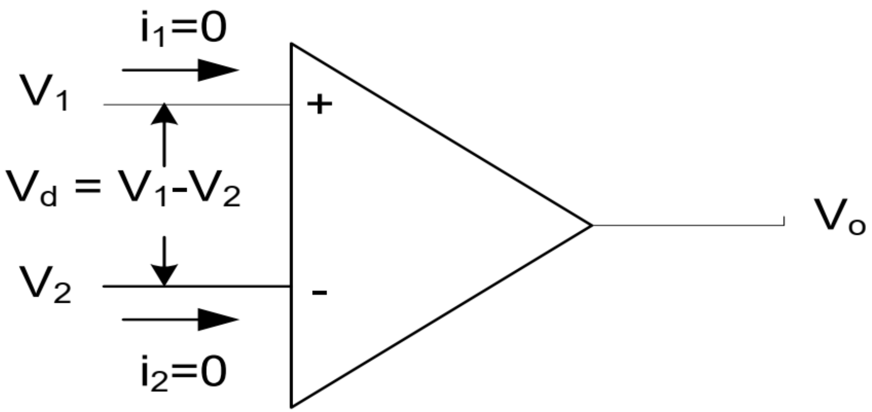
- An Op-Amp is a very high gain amplifier
- It has high input impedance (typically a few Mega ohm)

It has a low output impedance (less than 100Ω)

Characteristics of an ideal OP-Amp

- Input Resistance $R_i = \infty$
- Output Resistance $R_o = 0$
- Voltage Gain $A_v = \infty$
- Bandwidth = ∞ (i.e. can work over a wide range of frequencies)
- Perfect balance i.e $v_o = 0$ when $v_1 = v_2$
- Characteristics do not drift with temperature

Ideal Op-Amp analysis



i) $R_i = \infty$,

No current enters into op-amp

Voltage Gain $A_v = \infty$ or, $v_o / v_d = \infty$

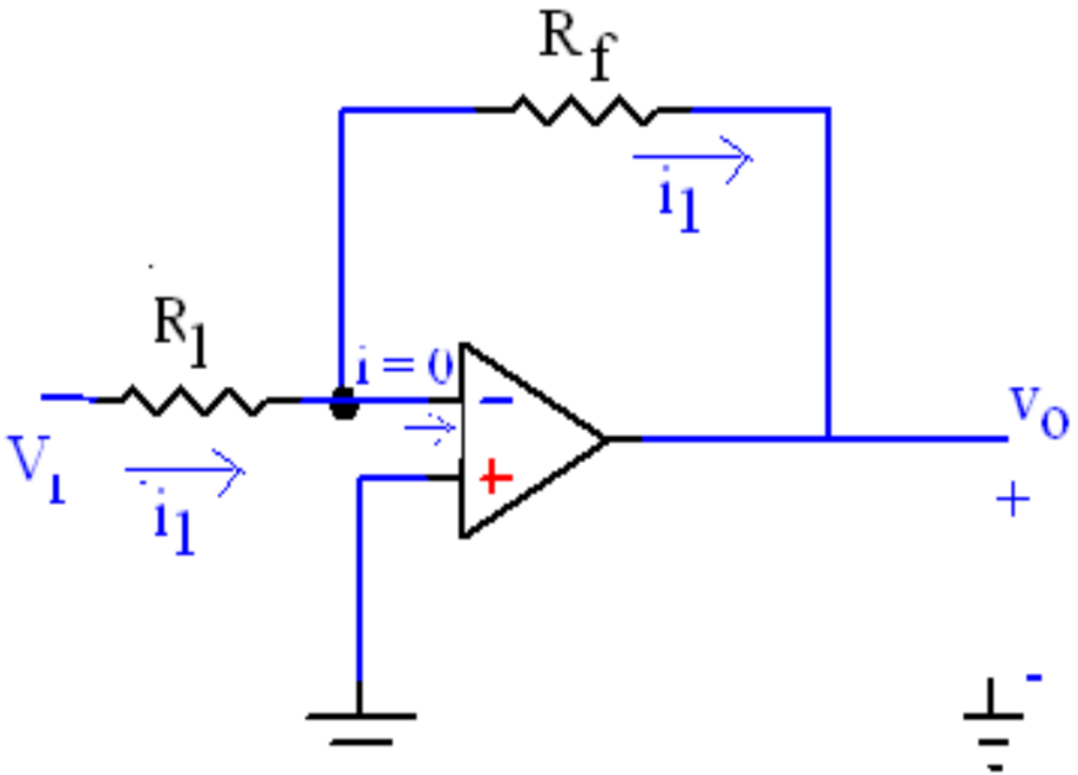
or, $v_d = v_o/\infty = 0$ [since v_o is finite]

Therefore, $v_1 - v_2 = 0$

or,

ii) $\mathbf{v_1 = v_2}$

1. Inverting Amplifier



The (-) terminal is effectively at ground. This is referred to as **"Virtual Ground"**

Using KVL,

$v_1 - i_1R_1 = 0$

$\Rightarrow \; i_1 = v_1/R_1$

&

$0 - i_1R_f - v_o = 0$

or, $v_o = -i_1R_f = -v_1R_f/R_1$

$v_0/v_1 = -R_f/R_1$

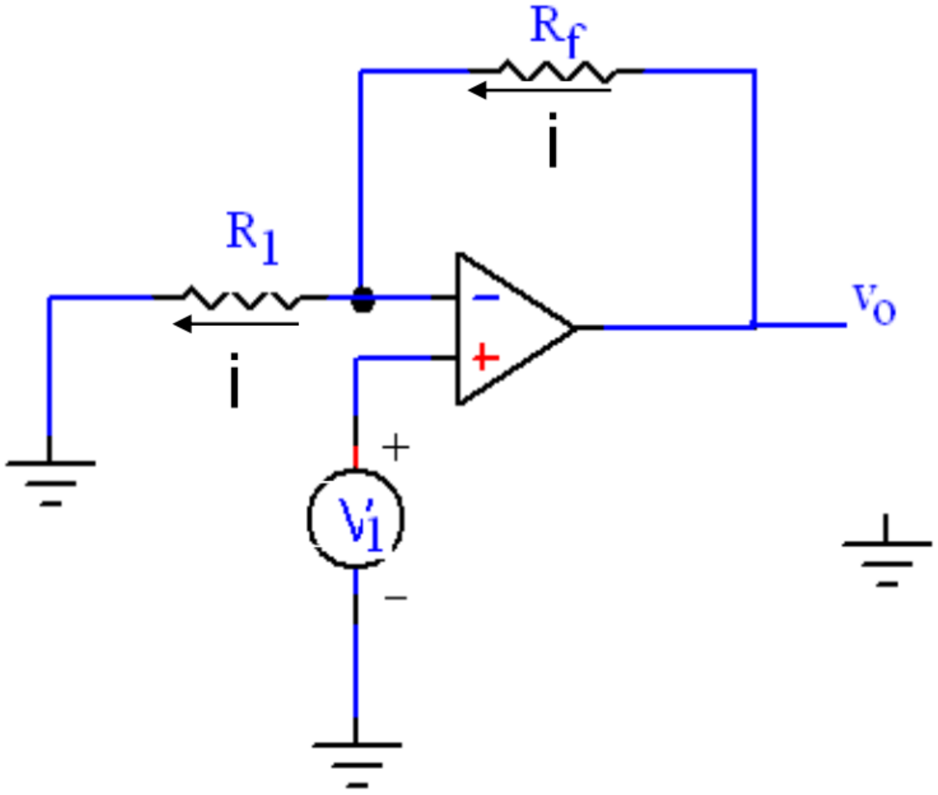
2. **Non Inverting Amplifier**

$$v_- = v_+ = v_1$$

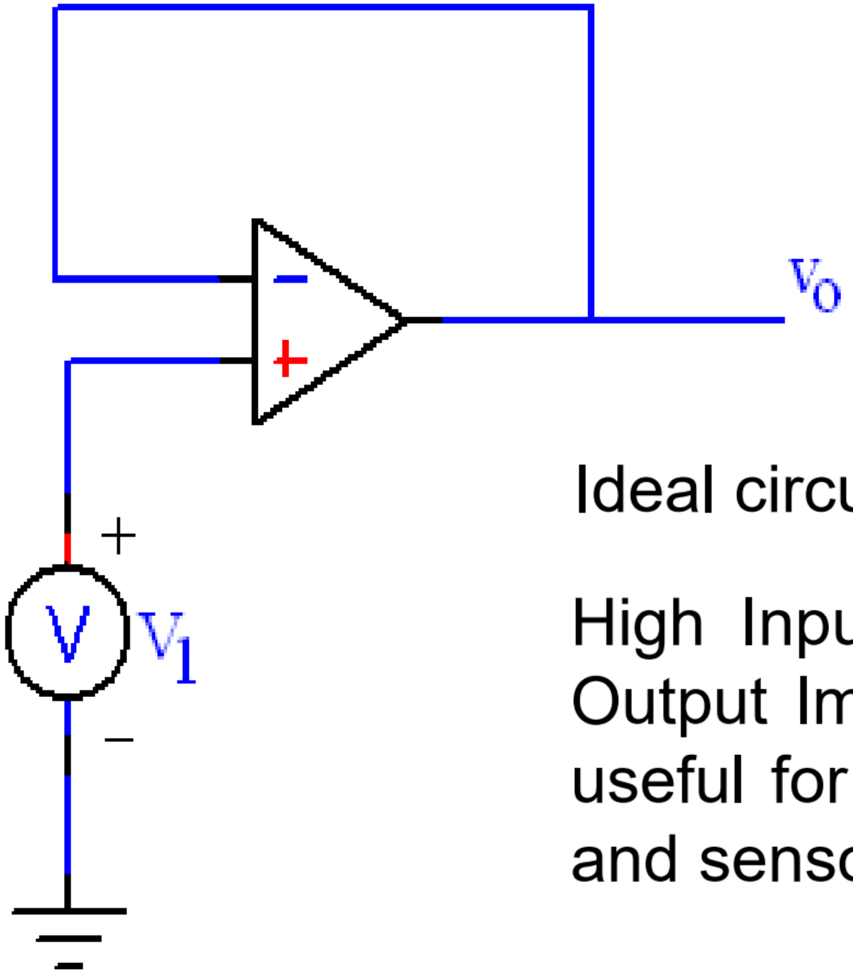
$$i = \frac{v_1}{R_1} = \frac{v_o - v_1}{R_f}$$

$$v_o = R_f\left[\frac{1}{R_1} + \frac{1}{R_f}\right]v_1$$

Gain $$\frac{v_o}{v_o} = \left(1 + \frac{R_f}{R_1}\right)$$



3. Voltage Follower



$$v_o = v_1$$

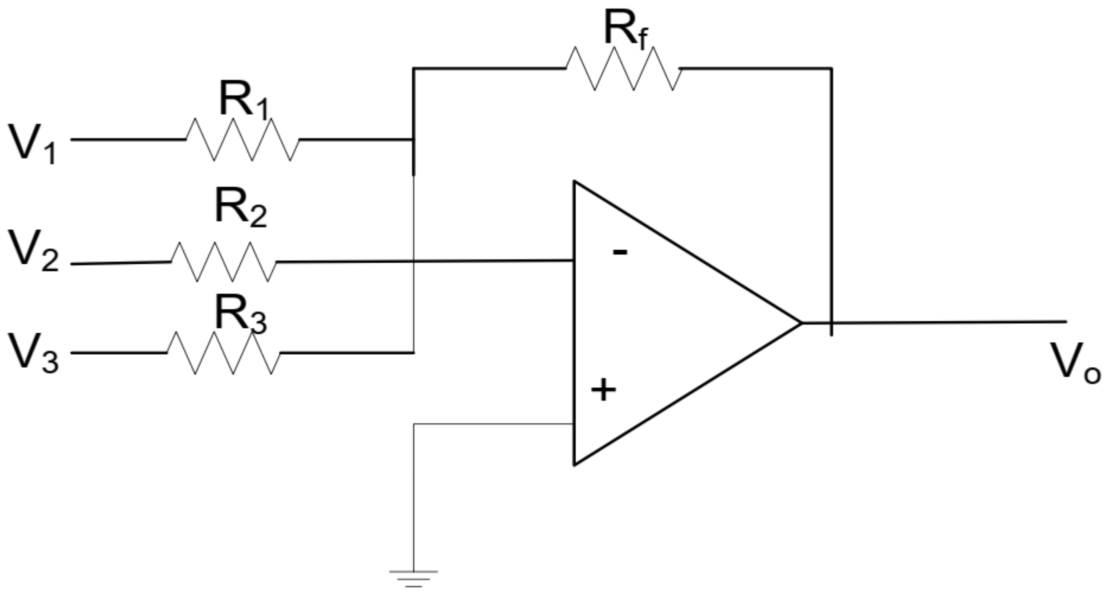
Ideal circuit to test Op-Amp

High Input Impedance and Low Output Impedance also makes it useful for interfacing transducers and sensors to other circuits

4. Summing Amplifier

Use superposition (i.e. consider one source at a time and add their respective outputs)

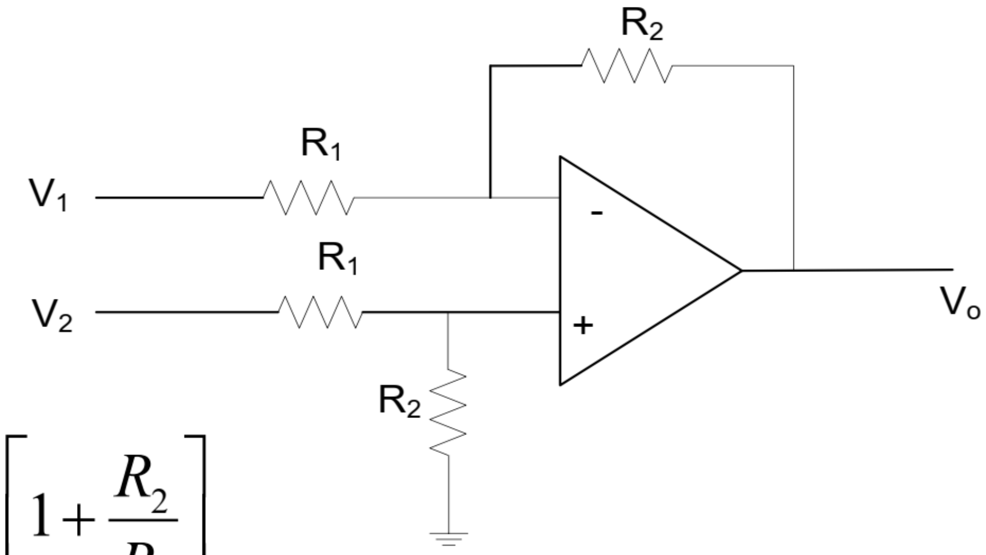
Can also be done directly



$$v_o = -\left[\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right]$$

Difference Amplifier (or Voltage Subtractor)

Use Superposition
(can also be done directly)



If v_2=0, $$v_{o1} = -\frac{R_2}{R_1} v_1$$

If v_1=0, $$v_{o2} = v_2 \left[\frac{R_2}{R_1 + R_2}\right]\left[1 + \frac{R_2}{R_1}\right]$$

$$= v_2 \frac{R_2}{R_1}$$

Therefore, $$v_o = v_{o1} + v_{o2} = \frac{R_2}{R_1}(v_2 - v_1)$$

Difference Gain = R_2/R_1

Instrumentation Amplifier

$$V_B = V_1 \quad V_C = V_2$$

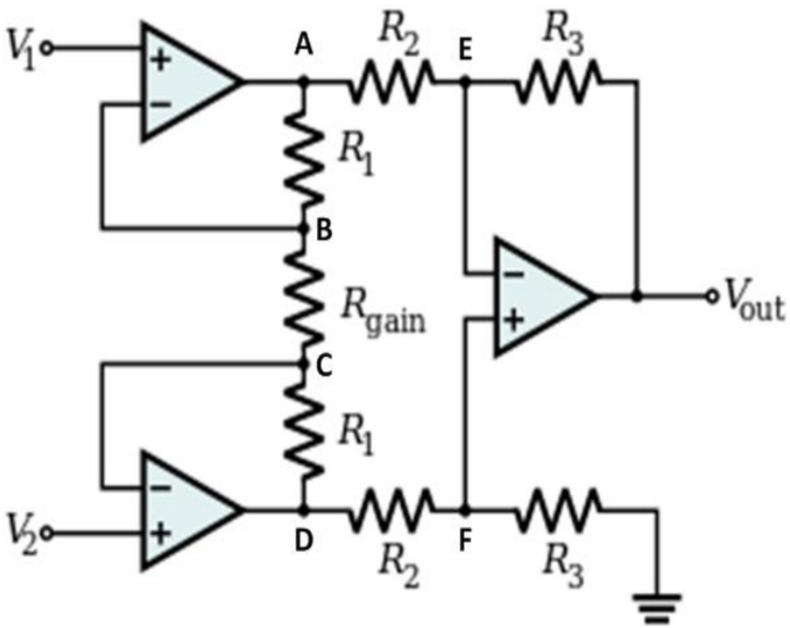
$$\frac{V_D - V_2}{R_1} = \frac{V_2 - V_1}{R_{gain}} \Rightarrow V_D = V_2 + \frac{R_1}{R_{gain}}(V_2 - V_1)$$

Similarly, $V_A = V_1 - \frac{R_1}{R_{gain}}(V_2 - V_1)$

$$V_F = \left(\frac{R_3}{R_2 + R_3}\right) V_D = V_E$$

$$\frac{V_{out} - V_E}{R_3} = \frac{V_E - V_A}{R_2} \Rightarrow V_{out} = \left(1 + \frac{R_3}{R_2}\right) V_E - \frac{R_3}{R_2} V_A$$

$$V_{out} = \frac{R_3}{R_2}(V_D - V_A) = \left(1 + \frac{2R_1}{R_{gain}}\right)\left(\frac{R_3}{R_2}\right)(V_2 - V_1)$$

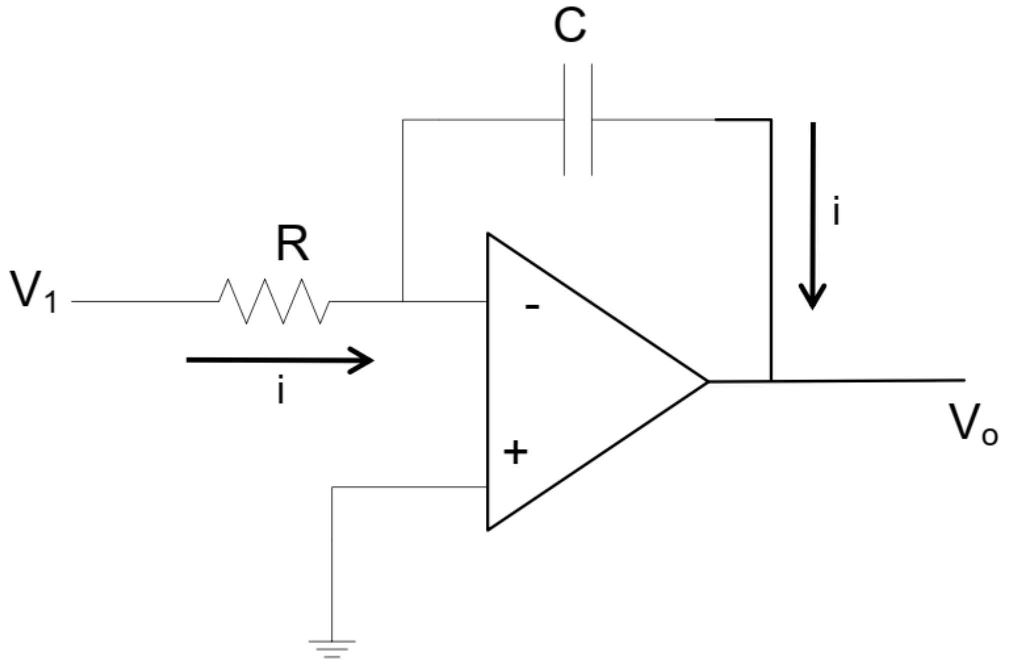


$$\frac{V_{out}}{V_2 - V_1} = \left(1 + \frac{2R_1}{R_{gain}}\right)\frac{R_3}{R_2}$$

Gain can be varied by just changing one resistor, R_{gain}

Integrator

$$i = \frac{v_1}{R} \qquad v_o = -\frac{1}{C}\int_0^t i\,dt \quad \text{with } v_o(0) = 0$$



Therefore,

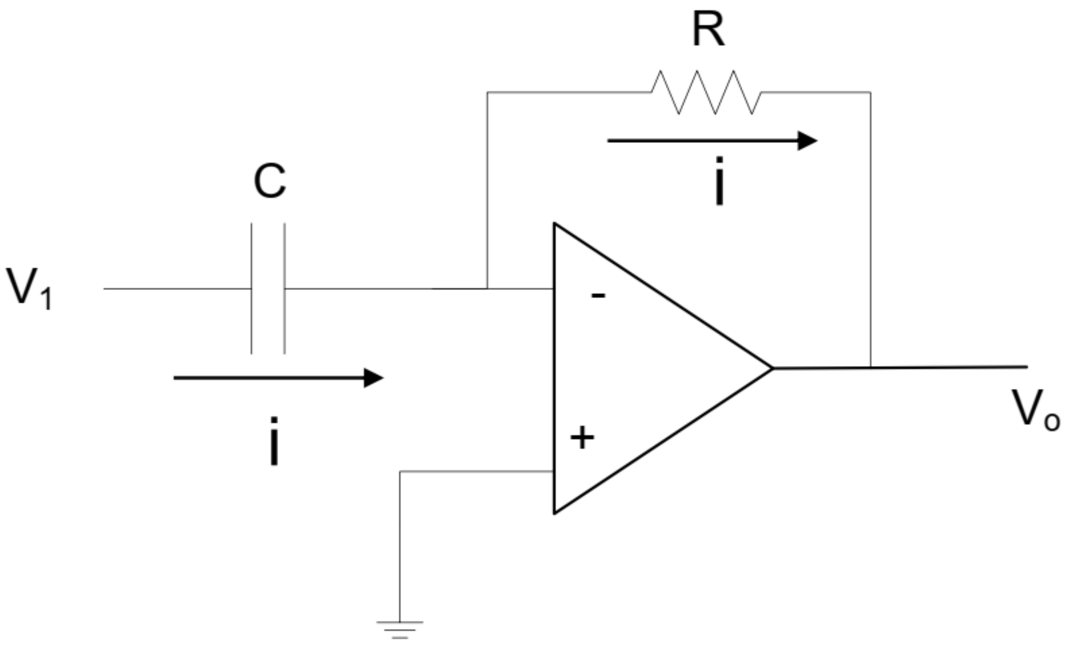
$$v_o = -\frac{1}{RC}\int_0^t v_1\,dt \quad \text{with } v_o(0) = 0$$

v_o

Differentiator

$$v_1 = \frac{1}{C}\int_0^t i\,dt \quad \text{with } v_1(0) = 0$$

$$v_o = -iR$$

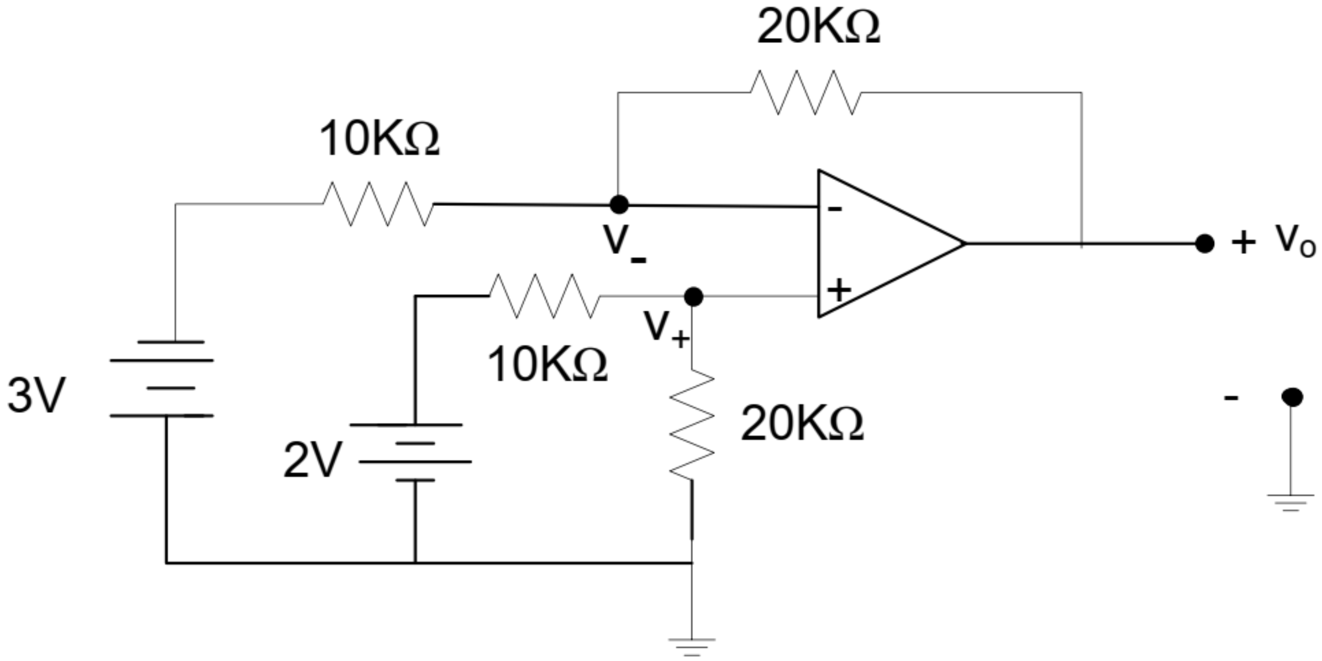


Therefore, $$v_1 = -\frac{1}{RC}\int_0^t v_o\,dt$$

or $$v_o = -RC\frac{dv_1}{dt}$$

Example

Find v_o in the given circuit.



$$v_- = v_+ = 2\times\frac{20}{30} = \frac{4}{3}$$

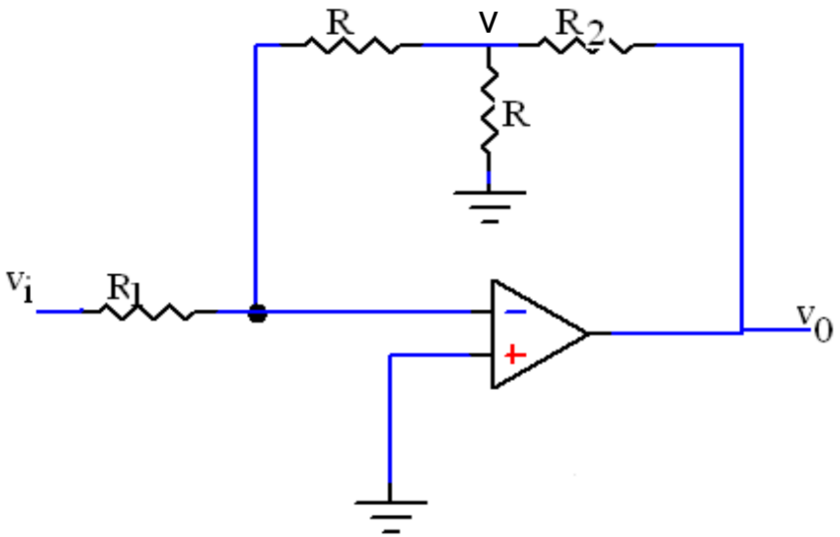
$$\frac{v_o - v_-}{20} = \frac{v_- - (-3)}{10}$$

$$\frac{v_o}{20} = v_-\left(\frac{1}{10} + \frac{1}{20}\right) + \frac{3}{10}$$

$$v_0 = 3v_- + 6 = 10 \text{ V}$$

Example

Find the gain v_o/v_i in the given circuit.



$$v_- = v_+ = 0 \quad \text{Virtual Ground at (-)}$$

$$\frac{v_i}{R_1} = -\frac{v}{R} \qquad \Rightarrow \qquad v = -\frac{R}{R_1}v_i$$

$$\frac{v_o - v}{R_2} = \frac{v}{R} + \frac{v}{R} = \frac{2v}{R}$$

$$v_o = v\left[1 + \frac{2R_2}{R}\right]$$

$$= -\left[\frac{R}{R_1}\right]\left[1 + \frac{2R_2}{R}\right]v_i$$

Gain $$A_V = \frac{v_o}{v_i} = -\left(\frac{R + 2R_2}{R_1}\right)$$

$$i_1 = \frac{v - v_{IN}}{2R} = \frac{v_O - v}{2R} \quad \Rightarrow \quad 2v - v_O = v_{IN}$$

$$i_2 = \frac{v}{2R} = \frac{v_L - v}{R} \quad \Rightarrow \quad v_L = \frac{3}{2}v$$

$$i_L = \frac{v_O - v_L}{R} - \frac{v_L - v}{R} = \frac{v_O - 3v + v}{R} = \frac{v_O - 2v}{R}$$

Therefore, $i_L = -\dfrac{v_{IN}}{R}$